Q1: Define *algorithmic bias* and provide two examples of how it manifests in AI systems.

Algorithmic bias occurs when systematic errors in machine learning produce unfair or discriminatory outcomes. Algorithmic bias can happen when the model is trained on limited data or skewed data, or when the data is incorrectly categorized or assessed, leading to poor outcomes. Additionally, programming errors can result in bias. For instance, if the input of the designer can introduce inaccuracies subject to their own consciousness or unconsciousness biases.

For instance, the use of AI during hiring can lock out many qualified candidates during recruitment process. Another example occurs when

Q2: Explain the difference between *transparency* and *explainability* in AI. Why are both important?

Transparency is the extent to which information about AI system's design, operations, and decision-making processes are open, accessible, and understandable to stakeholders while *explainability* is the ability to provide understandable justification for system's decisions, outputs, and behaviors.

Both transparency and explainability are important since they focus on different elements in AI ethics, responsible AI, and AI governance. Transparency promotes trust in the system while explainability focuses on establishing trust on specific outputs.

Q3: How does GDPR (General Data Protection Regulation) impact AI development in the EU?

GDPR has impacted how data is collected and processed. In case of AI, companies are required to collect and process necessary data that helps them achieve their intended purpose. For instance, if a company collects data for model training, it should limit to that purpose, achieving data minimization and limitation.

Companies need to anonymize data so that safeguard personal data and enhance privacy. AI systems processing data should not be able to identify individual's personal data. Additionally, GDPR prohibits companies from storing data for longer periods. Thus, companies should always maintain up-to-date records.

Match the following principles to their definitions:

A) Justice: *Fair distribution of AI benefits and risks.*

B) Non-maleficence: *Ensuring AI does not harm individuals or society.*

C) Autonomy: *Respecting users' right to control their data and decisions.*

D) Sustainability: *Designing AI to be environmentally friendly.*

Case Study Analysis

Case 1: Biased Hiring Tool

- **Scenario**: Amazon's AI recruiting tool penalized female candidates.
- **Tasks**:

1. Identify the source of bias (e.g., training data, model design).

The bias arose when the data fed to the model was from males with at least 10 years' experience. This led to preference of male candidates in the tech industry downgrading female candidates for technical jobs.

2. Propose three fixes to make the tool fairer.

   a. Ensure the algorithm is fair, transparent, and explainable.
   b. Re-train the algorithm with balanced data i.e. male and female resumes.
   c. Embrace regular auditing and monitoring of the model after deployment.

3. Suggest metrics to evaluate fairness post-correction.

   - Perform demographic parity to ensure different groups are equally represented.
   - Ensure predictive parity which ensures predictive positive outcome has the same precision across different groups.
   - Continuous monitoring and auditing of the model.
   - Evaluate using multiple metrics.

Case 2: Facial Recognition in Policing

- **Scenario**: A facial recognition system misidentifies minorities at higher rates.
- **Tasks**:

1. Discuss ethical risks (e.g., wrongful arrests, privacy violations).

   - Violation of Human Rights: Individuals can face wrongful arrests due to wrong identification. The individuals may need to prove their innocence since the technology identifies them to be guilty, violating human rights.

- Indiscriminate surveillance: People are unable to give consent thus authorities can use the facial recognition system to track individuals. This leads to loss of anonymity and continuous monitoring.
- Racial biases and discrimination: The system can be biased against immigrants and people of different races or the vulnerable in the community leading to arrests and trial of wrong individuals.
- Data Privacy: There are an unauthorized data collection, surveillance, and misuse of data.

2. Recommend policies for responsible deployment.

   - Ensure models are trained on diverse datasets, conduct continuous audits, and mitigate bias testing.
   - Standardize consent processes, regulate dataset sourcing, and ensure disclosure policies.
   - Encrypt facial data, enforce breach disclosure, and establish stronger biometric laws.